

UGROP - Wayne State

Final Report

Understanding and Improving Truthfulness in LLM's

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Research Problem

Currently many large language models (LLMs) perform better on most tasks when trained on larger amounts of data [8]. However, it has been shown that an inverse scaling problem exists in certain tasks such as adversarial truthfulness [11]. Adversarial truthfulness exposes the difficulty LLMs have with questions which are more contentious, conspiratorial, or for which there is a greater mix of possible answers in the dataset. The TruthfulQA dataset was designed to exploit misconceptions learned by LLMs from training data, and larger models were shown to perform worse on this dataset by (Lin et al, 2023) [11]. Some baselines have been established for ChatGPT-3.5-Turbo on this dataset [8]. And several finetuning methods have been used to improve performance [9], [2], with a few being tested on LLaMA [10], [7], as well as in-context learning [15].

In this project we aim to better understand the important components of truthfulness; first by comparing existing methods of finetuning such as using a collection of human evaluated true question and answer pairs as training examples – and then testing some of the hypotheses stemming from these methods. It has been hypothesized that truthful agents represent a 'truthful persona', the features of which are learned by LLMs through finetuning [6]. This has been previously explored in a tangential manner, where linguistic properties of a truthful response are used [9]. We hoped to further explore this hypothesis by expanding on the methods used in (Joshi et al, 2023) and (Lee et al 2023). However, we see later in our work that the SVM methods used for distinguishing these features were affected greatly by a class imbalance in the SVM test data. We hypothesize that modeling a persona is effective at improving the truthfulness of agents, which we see in the performance of our fictionally tuned model. We also find that modeling untruthful personas is separable from the actual facts these training examples contain, as seen in the poor performance of our model tuned to bad-faith or at least confused statements from the internet. This work helps to expand our understanding of the key components of truthfulness and tease apart pure fact-recall from these other more subtle hypotheses.

Additionally, there are challenges to the safety and strength of a model's truthful alignment [17]. We proposed that methods from (Joshi et al, 2023) and (Lee et al 2023) can be used to mask an untruthful agent as truthful. Using hand-coded linguistic features from Lee and personas from Joshi, we planned to finetune LLMs adversarially. Due to problems with our SVM having a serious prediction imbalance however, we choose to forgo the extraction of features and instead introduce another fine tuning dataset built from covid information falsities hoping to understand the modulus between fiction with a truthful persona, and fiction proper.

Fine-Tuning: an Overview

Parameter efficient fine tuning (PEFT) is a method of altering an already trained model by training a small set of parameters in an attempt to improve the performance of a model on some given task or domain [12]. This method allows for the tuning of specific components of

a model without retraining the entire network, saving computational resources/time. Another method attempting to achieve a similar result is in-context learning, whereby a few training examples are given to the model directly with the input. It has been established previously in the literature that PEFT is more effective than in-context learning for some tasks. We might wonder how fine tuning enables the improvement on any tasks in the first place, and whether it has the potential to disrupt the distribution a model has previously learned.

Prakash et al address the problem of explaining the influences of fine tuning in a recent pre-print, using the specific task of entity tracking to compare model circuits used in this task before and after fine-tuning [14]. They identify the output of a group of learning heads they call Value Fetcher heads as encoding a more enhanced representation of the object than in the untrained model, as well as noting the persistence of the original circuits involved. To relate this to our current work, we might suggest exploring how the original circuits involved in adversarial question answering are changed under fine-tuning – whether it improved or impaired the models performance on the task.

In a recent paper, Tian et al some particular methods for achieving factuality in biography-style long form writing [16]. To improve the performance of their fine tuning activity, they construct a method for estimating the truthfulness of model responses using Llama3 as well as Chat-GPT. By prompting a model multiple times with the same prompt, and then selecting the most truthful responses to reward using PEFT. This method shows how we might improve the factuality of a model, and in the future could potentially be used for adversarial facts as well.

Linguistic Properties of a Truthful Response

We based our experiments on the hypothesis posed by [6] out of UPenn postulating the existence of a truthful persona, as well as [9] who uses linguistic features to distinguish truthful responses. In the paper by Joshi et al, they postulate that the introduction of sets of true facts used as fine tuning examples being sufficient to improve model performance on the TQA dataset supports the existence of the model of a truthful persona. We reproduce this later using true facts from an alternative data source for fine tuning, and then explore this further by introducing a dataset of 'fictional facts' in which facts only true in a fictional domain are used to tune a model and ultimately test on the TQA benchmark. Additionally, they are able to show that the truthfulness of a model response can be inferred by model activations before the answer is generated.

Joshi shows, like we later do, that fine tuning on untruthful question answer pairs does hinder the model's performance on TQA [6]. Though we reproduce this in our experiments, we also further corroborate the persona hypothesis by showing how the factuality of a statement is not the same as it's truthfulness.

We also explored the work of Lee, both looking into the truthful persona hypothesis and applying their linguistic features python package to our own datasets in an attempt to curate datasets with truthful features. This model uses 220 linguistic features such as adjective count and variation as input, correlating to a single model response. The SVM model attempts to predict if the model response is true or not before exposure to the human generate class label. On reproducing the davinci model found in the source code, we found a class imbalance in the test data. For our reproduced davinci model, we predicted a 0 class label for every test example, which still resulted in an accuracy of 76.5%. This reproduction seems to suggest that the recall for this model is poor, and may be conflating the effect of linguistic features have on this actual truthfulness of model responses. In light of this we forgo the use of linguistic features in our final fine tuning dataset curation. To keep our investigation into this hypothesis going we supplement our 'fictional facts' dataset with a dataset of poorly written falsities in order to distinguish the tuning effect between the existence of non-facts and the absence of

quality linguistic features.

Experiment Setup

Initially we use batch processing to send the TQA questions directly to gpt-turbo to establish a baseline for our evaluation methods. We collect model responses in the form of a jsonl and then parse the file for raw model responses. We then format another file which pairs the responses of the model to the provided 'correct' responses from the TQA dataset. To avoid human evaluation, this prompt asks if the two responses contain the same information. This is how the evaluation takes place for each tuned model as well. We expect this method to result in lower accuracy, as it does not account for divergent but still true responses. The baseline accuracy established by this method was 47.9%.

In order to create individually custom tuned models from chat-gpt-3.5-turbo we utilize the open ai api and batch processing dashboard. We first collect a training example set of around 100 question and answer pairs for each model, using the code found on our github to format jsonl files matching the requirements for tuning and batch processing provided by openai [13]. By sending a fine tuning jsonl file we are able to create a specific model hosted on openai's system. Upon fine tuning a model we can then perform the evaluation detailed in the previous paragraph.

Fine Tuning

For our first fine tuning experiment we collect true facts in a question/answer format from the MS MARCO dataset [3]. Pulling 110 examples from the dataset, we then format training examples and create a fine tuning model using the openai api. The topic of these questions are basic facts, such as "Who was the duke in the battle of Hastings?" and "The hardest problems in NP can be analogously written as what class of problems?". It should be noted that the answers to these questions are brief few-word answers. After testing our tuned model on the TQA dataset and automatic evaluation, we received an accuracy of 45%.

For our subsequent tuning experiment we look to a fictional domain of true facts, in this case the fictional universe of Game of Thrones (GoT). To curate this dataset, we scraped a well written wiki page which is filled with information from the GoT universe and then prompted chat-gpt to create questions based on this data [1]. In our prompt we instruct the model to use only quotes from the original webpage as the material for an answer, a prompt which it complied to very well. The adherence to these instructions is important to our result, so we manually check to ensure the resulting 100 examples do in fact contain text directly from the fictional truth teller that is the GoT wiki page. An example question and answer pair from this dataset: "Which languages can Daenerys speak?", "Beside the Common Tongue, Daenerys speaks the bastard Valyrian of the Free Cities with a Tyroshi accent, and High Valyrian." We note the quality of punctuation and the presence of several fictional proper nouns.

Upon Compiling our question answer pairs, we then format them into a tuning file and create more models. For this experiment we performed a grid search on the tuning parameters in an attempt to maximize the performance of this tuning method. We use the openai api to create several models, each with differing tuning parameters, and then test each model individually on the TQA dataset. We find great use for our automatic evaluation method here. Below we note the performance of each parameter instance. 'd' denotes the default parameter setting. We also note the qualitative appearance of the tuning loss under 'Convergence'.

We see an automatically evaluated accuracy of up to 47.1%, closely mirroring the baseline performance of the model with no tuning.

Epochs	Batch	L-rate	Loss	Convergence	Accuracy
d	d	d	1.4	No Convergence	0.3594
1	1	0.1	1.47	Convergence	0.4560
1	4	0.1	2.4	Strong Convergence	0.4450
1	5	0.1	2.4	No Convergence	0.4609
5	20	0.3	1.69	Convergence	0.4707
10	10	0.5	1.35	Medium Convergence	0.4022

Table 1: Fictional Tuning All Results

Liar Tuning

Previously in our fictional tuning model we used a dataset of facts which are in fact not true. Despite this we received a promising accuracy from our tuning. In order to explore the hypothesis of Joshi further, we wish to create a differentiating model which is trained on falsities of a linguistically less trustworthy nature. To do this we identify a dataset from Das containing true and false statements made online regarding the Covid-19 pandemic [5]. We first filtered the data to include only statements labeled false by human judgment, then truncated it to include 100 examples. We then followed our previous procedure by prompting chat-gpt4 to create questions using this dataset, again prompting the ai to construct questions which use only the source answer text.

Then we collect our question and answer pairs into a jsonl file to submit to the fine tuning endpoint, creating a model tuned on false and unorganized question/answer pairs. Here is an example of one of these pairs: "Which organization holds the patent for the coronavirus?", "The Gates Foundation has the patent for this coronavirus". We see a statement written in plain english but which contains a false truth. Though this answer is false, it is similar to our fictional tuning set as such.

Upon testing this model using the TQA dataset, we see a dismal accuracy of 26%. This result differs greatly from any of our previous results.

Results/Conclusion

In this paper we explored the impact of fine tuning on large language models (LLMs) using the TQA dataset [11], delving into current research on the topic and focusing on the influence of both true, fictional, and false information on model performance. Our experiments, inspired by the Joshi hypothesis that LLMs can be positively influenced by a 'truthful personal' through fine-tuning, provided insight into the influence fact-recall and the underlying linguistic properties impacting a model's truthfulness during the fine tuning process.

Our first experiments demonstrated that fine-tuning on truthful data from various domains, including both factual and fictional sources, can enhance model performance on truthfulness benchmarks like TruthfulQA. Notably, our "fictional tuning" experiment, which used facts from the fictional Game of Thrones universe, achieved accuracy levels comparable to our baseline. This suggests that even well structured fictional truths can contribute to a model's truthfulness.

The challenges we encountered with class imbalance in our SVM-based linguistic feature reproduction forced us to alter our plans slightly and focus on personas from an alternate direction, and we were unable to test whether or not a model could be injected with a truthful persona which more often than not relays false information. While the SVM's poor recall and the resulting class imbalance limited the effectiveness of our linguistic feature-based approach, it showed the difficulty of isolating truthfulness from mere linguistic patterns. We suggest, like the original authors do, that for future work the SVM in question is trained on a much larger dataset. The human evaluation of the truthfulness of those features greatly limited the amount

of data available for training.

Our attempts to fine-tune models on datasets containing false information in the Covid-19 misinformation dataset resulted in significantly lower performance. This outcome highlights the importance of the quality of writing and the nature of the speakers voice in the data used in fine-tuning and suggests that models tuned on data with a less-than-truthful persona are less capable of maintaining truthfulness, especially in the face of adversarial tasks.

Overall, our findings suggest that while fine-tuning can indeed improve a model’s alignment with truthfulness, the choice of training data is crucial. Models tuned on coherent and truthful data, even from fictional domains, tend to perform better than those exposed to disorganized or false information. This work contributes to a deeper understanding of the components of truthfulness in LLMs and sets the stage for future research aimed at refining fine-tuning methodologies to better prevent ai agents from acting untruthfully.

Future Work

This study has illuminated several potential avenues for further research, particularly in refining fine-tuning methodologies to enhance the truthfulness of large language models (LLMs). One promising direction is to continue exploring the "truthful persona" hypothesis, particularly by addressing the challenges faced in the SVM-based linguistic feature reproduction. Future work could involve training the SVM on a larger, more diverse dataset. With a focus on overcoming the issues that limited our current investigation. Additionally, expanding the dataset used for training could help improve the recall and accuracy of identifying truthful responses based on linguistic patterns, offering a more robust evaluation of the truthfulness hypothesis.

Another area of interest lies in the potential for adversarial fine-tuning, where models could be trained on datasets containing deliberate misinformation or adversarially framed facts. This would help test the limits of a model’s ability to maintain truthfulness under challenging conditions and further our understanding of how fine-tuning can influence model behavior in adversarial contexts. Moreover, fine-tuning on a broader array of fictional domains, beyond the Game of Thrones universe, could offer insights into how different types of fictional truths impact a model’s performance on truthfulness benchmarks. The addition of some linguistic feature extraction and modeling would allow us to construct more diverse and precise datasets.

The code for this project can be found on github [4].

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